

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 2

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GitHub Repository Link: <https://github.com/miradonkoh/lab-2-predictive-analytics.git>

Date Submitted: 3rd July 2025

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	yes
Estimated percentage of the work influenced by AI	15-20%
Did you attach evidence of AI use where applicable?	yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Chat gpt	To debug my errors	Id take screenshots		I analyzed the debugged code, and made it suit mine.

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft		
Prompt history		
Screenshot of AI interaction		
Exported AI conversation		
AI-generated code snippet		
Revised version showing student input		

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature:Samira Donkoh

Date:3rd July 2026

```
i got this error?but i defined y a;ready# TODO: Separate features (X)
and target (y = tip_amount)
X=taxi.drop("tip_amount",axis=1)
Y=taxi["tip_amount"]
# TODO: Split into train / validation / test (a common choice is
60/20/20).
from sklearn.model_selection import train_test_split
# First split off the test set, then split the remainder into train +
validation.
# Use random_state=RANDOM_STATE.
Xtrain,Xtemp,ytrain,ytemp=train_test_split(X,y,test_size=0.4,random_stat
```

Show more ▾



The error is very literal:

```
NameError: name 'y' is not defined
```

Python is telling you it can't find a variable called `y`.

What went wrong

You defined:

Python



Run

```
Y = taxi["tip_amount"]
```



But later you used:

+ Ask anything



```
im still getting the rroer # TODO: Separate features (X) and target (y =
tip_amount)
X=taxi.drop("tip_amount",axis=1)
Y=taxi["tip_amount"]
# TODO: Split into train / validation / test (a common choice is
60/20/20).
from sklearn.model_selection import train_test_split
# First split off the test set, then split the remainder into train +
validation.
# Use random_state=RANDOM_STATE.
RANDOM_STATE=42
Show more ▾
```

You're *very close* — the problem is still just a **variable mismatch**.

✖ What's wrong in your code

You defined:

Python

```
Y = taxi["tip_amount"]
```



Run

But you used:

Python

```
train_test_split(X, y, ...)
```



Run

+ Ask anything



I GOT THIS error too ValueError Traceback (most recent call last)
/tmp/ipykernel_8452/1078790114.py in <cell line: 0>()
13 from sklearn.preprocessing import StandardScaler
14 scaler=StandardScaler()
---> 15 Xtrainscaled=scaler.fit_transform(Xtrain)
16 Xvalscaled=scaler.transform(Xval)
17 Xtestscaled=scaler.transform(Xtest)

8 frames
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py in
[Show more](#) ▾

Good — this error is **different and very common in real ML preprocessing**.

✗ What it means

Input X contains infinity or a value too large for dtype('float64')

Your `Xtrain` contains:

- `inf` (infinity), or
- `-inf`, or
- extremely large values

👉 `StandardScaler` cannot handle those.



+ Ask anything



it underlined this row though `Xtrain_scaled=scaler.fit_transform(Xtrain)`

If that line is underlined, it usually means **the problem is still inside `Xtrain`**, not the scaler itself.

Given your earlier error, there are only 2 realistic causes left:

✅ **Cause 1: `Xtrain` still contains `NaN` or `inf`**

Even one remaining bad value will break `StandardScaler`.

🔧 **Quick fix (most reliable, minimal change)**

Run this **immediately** before scaling:

Python

```
import numpy as np

Xtrain = Xtrain.replace([np.inf, -np.inf], np.nan)
Xval = Xval.replace([np.inf, -np.inf], np.nan)
Xtest = Xtest.replace([np.inf, -np.inf], np.nan)

Xtrain = Xtrain.fillna(Xtrain.median(numeric_only=True))
Xval = Xval.fillna(Xtrain.median(numeric_only=True))
Xtest = Xtest.fillna(Xtrain.median(numeric_only=True))
```

+ Ask anything

